Available online at www.sciencedirect.com
SCIENCE DIRECT®
Carbon 43 (2005) 1239–1245
CARBON
www.elsevier.com/locate/carbon

Mechanical behavior and microstructure of cement composites incorporating surface-treated multi-walled carbon nanotubes

Geng Ying Li [a,b,*], Pei Ming Wang [b], Xiaohua Zhao [a]

[a] *Department of Civil Engineering, Shantou University, Shantou 515063, China*
[b] *State Key Laboratory of Concrete Material Research, Tongji University, Shanghai 200092, China*

Received 20 May 2004; accepted 21 December 2004

Abstract

Multi-walled carbon nanotubes after modified by using a H_2SO_4 and HNO_3 mixture solution were added to cement matrix composites. The mechanical properties of the newly formulated composites were analyzed, and the results show that the treated nanotubes can improve the flexural strength, compressive strength, and failure strain of cement matrix composites. The porosity and pore size distribution of the composites were determined by using Mercury intrusion porosimetry, and it is observed that the addition of carbon nanotubes can fine the pore size distribution and decrease porosity. The phase composition was characterized with Fourier transform infrared spectroscopy. It is found that there are interfacial interactions between carbon nanotubes and the hydrations (such as C–S–H and calcium hydroxide) of cement, which will produce a high bonding strength between the reinforcement and cement matrix. The mineralogy and microstructure were analyzed by using scanning electron microscope. It is shown that carbon nanotubes act as bridges across cracks and voids, which guarantees the load-transfer in case of tension.

Keywords: Carbon nanotubes; Infrared spectroscopy; Scanning electron microscopy; Pore size distribution

1. Introduction

Carbon nanotubes (CNTs) have an ideal structure formed by carbon atoms with one dimension [1]. They are seamless tubes of graphite sheets with nano-sized diameter. Depending on their precise structure, carbon nanotubes can be either metallic conductors or semiconductors. Carbon nanotubes have a very high theory strength, 100 times more than that of steel, although their specific gravity is only one sixth that of the latter [2]. Moreover, carbon nanotubes have an elastic strain capacity of 12%, 60 times higher than that of steel [3]. In addition, the electrical behavior of carbon nanotubes can vary with stress. According to Tombler [4], the resistance of carbon nanotubes will increase 100 times when strain changes from 0.0% to 3.2%.

Since invented by Iijima [5] in 1991, carbon nanotubes have been widely used in fields such as electronic materials, biological technology, chemistry, and multifunctional composites. It was reported that the addition of carbon nanotubes could greatly improve the mechanical behaviors of polymer-based composites. Studies showed that more than 500 MPa of stress could be transferred through the interface between polymer and a carbon nanotube, 10 times higher than that between polymer and a carbon fiber [6,7]. It was also reported

* Corresponding author. Tel.: +86 754 836 7533; fax: +86 754 290 2005.
E-mail address: gyli@stu.edu.cn (G.Y. Li).

0008-6223/$ - see front matter © 2005 Elsevier Ltd. All rights reserved.
doi:10.1016/j.carbon.2004.12.017

that the electrical conductivity of polymer would increase about 10^8 times, when carbon nanotubes were added in the amount of 7–11% (by weight mass). Nowadays, carbon nanotubes have also been used to reinforce many kinds of materials including metals [8–11] and ceramics [12–15].

Cement is a construction material commonly used due to its low cost and high compressive strength, and to enhance its performance has been a concern of the research community. Recently, carbon filaments and conventional carbon fibers have been used as fillers in cement matrix composites to improve both mechanical and electrical behaviors of the material, as well as the electromechanical and electromagnetic behaviors [16–19]. Results showed Carbon filaments are superior to conventional carbon fibers (discontinuous) as filler in providing cement–matrix composites for electromagnetic performance, but are inferior to the fibers as filler for composites for electrical and mechanical performance.

Having the best performance among carbon materials, it is expected that carbon nanotubes can remarkably enhance the performance of cement composites. When carbon nanotubes are used to form advanced composites, a crucial problem arises, namely the interfacial bonding of nanotubes to cement matrix composites. According to Chung [16], surface treatment of carbon by using ozone gas (0.3 vol% in air, 160 °C, 10 min) can increase the tensile strength and compressive strength. Presently researches show that carbon nanotubes treated by using a H_2SO_4 and HNO_3 mixture solution may lead to the formation of carboxyl acid groups on their surfaces [20,21]. The presence of carboxylic acid groups will induce a series of chemical reactions along the interface, and therefore enhance the reinforcement efficiency.

In this paper, the acidic treated carbon nanotubes are added to cement, and both the mechanical behavior and microstructure attributes of the newly formulated composite are studied. The phase composition was characterized with Fourier transform infrared spectroscopy, while the mineralogy and microstructure were studied by using Scanning electron microscope (SEM). The porosity and pore size distribution were determined by using Mercury intrusion porosimetry (MIP). Indeed, the results show that the addition of carbon nanotubes to cement can improve both its mechanical behavior and microstructure. For comparison, the conventional carbon fiber-reinforced cement-matrix composites were also studied.

2. Experimental program

2.1. Materials

The cementitious material used in the test was ordinary Portland cement. Its chemical composition and physical properties are shown in Table 1. The fine aggregate was natural standard sand (Chinese standard) with a specific gravity of 2.62.

The purified multi-walled carbon nanotubes, provided by Shenzhen NANO-Technology Company (Limited) in China, were used. Their properties are given in Table 2, and their morphology and microstructure are shown in Fig. 2.

The carbon fibers were isotropic pitch based, unsized, and of nominal length 6 mm and diameter 10–14 μm, as obtained from East-Asia Company of Shengyang, China. The properties of carbon fibers are presented in Table 3.

2.2. Carboxylation of nanotubes

The carbon nano-tubes were all treated firstly to attach carboxylic acid to their surfaces, and the procedure was in the following: one hundred grams of nano-tubes were taken and added to one thousand millilitres of the mixed solution of sulfuric acid and nitric acid (3:1 by volume, respectively). The mixture was sonicated in a basin for 3 h at ambient temperature. Then the mixture

Table 2
Properties of carbon nanotubes (CNTs)

External diameter	10–30 nm
Length	0.5–500 μm
Ash	0.2%
Purity	95
Special surface area	40–300 m^2/g
Amorphous carbon	3%
Electric conductivity	10^2–10^{-4} s/cm

Table 3
Properties of carbon fibers

Filament diameter	10–14 μm
Nominal length	6 mm
Tensile strength	⩾241 MPa
Tensile modulus	27.5–41.6 GPa
Electrical resistivity	$15–30 \times 10^{-3}$ Ωcm
Carbon content	⩾90 wt.%

Table 1
Chemical and physical properties of cement

SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	SO_3	LOI	Specific surface, Blaine, $m^2 kg^{-1}$	Compressive strength, 28-day, MPa
20.6	4.0	3.1	62.8	2.6	3.1	1.8	392	56.7

was diluted with distilled water in a ratio of 1:5 by volume. After holding still for 24 h, the top part of the diluted solution was carefully removed. The remained solution was diluted further. Repeated four times, the mixture was filtered with a filter, and washed with water until no residual acid was present. Thereafter, the treated nanotubes were characterized by Fourier transform infrared spectroscopy (FT-IR).

2.3. Preparation of specimens

Three mixes of cement paste (namely plain cement paste, cement paste containing untreated carbon fibers, cement paste containing surface-treated carbon nanotubes) were prepared for FT-IR testing. The water/cement ratio of these cement pastes was 0.45. In order to reduce the interfered of unhydrated cement, those cement pastes were analyzed after 90 days curing. Also, in order to enhance the FT-IR spectrum peaks of the cement pastes containing carbon nanotubes and fibers, carbon nanotubes and fibers were added in the amount of 2% by weight of cement.

Three mixes of cement mortar were prepared for SEM and MIP testing: (i) control Portland cement composites (PCC), for which the ratio of water/cement/sand was 0.45:1:1.5. (ii) Cement mortar containing untreated carbon fibers (PCCF). The ratio of water/cement/sand was 0.45:1:1.5, and carbon fibers were added in the amount of 0.5% by weight of cement. To help the fiber dispersion, methylcellulose in the amount of 0.4% by weight of cement was added, and a defoamer in the amount of 0.13 vol% was used with methylcellulose, (iii) Cement mortar containing treated carbon nanotubes (PCNT). The ratio of water/cement/sand was 0.45:1:1.5, and treated carbon nanotubes were added in the amount of 0.5% by weight of cement.

A rotary mixer with a flat beater was used for mixing. In cases of control Portland cement composites, cement, sand (if needed), water and water-reducing agent were all mixed in the mixer for 5 min. In cases of cement paste and mortar containing carbon fibers, methylcellulose was firstly dissolved in water, and then both the fibers and the defoamer were added. The solution was stirred by hand till the fibers were scattered (about 2 min). When finished, the solution, cement, sand (if needed) and water-reducing agent were mixed in the mixer for about 5 min. In cases of cement paste and mortar containing treated nanotubes, cement, treated nanotubes and sand (if needed) were mixed in the mixer for 3 min. Then, water and water-reducing agent were added. The mixture was mixed for another 5 min.

After pouring the mixes (both cement paste and cement mortar) into oiled molds ($40 \times 40 \times 160$ mm), an electric vibrator was used to ensure good compaction. The specimens were then surface-smoothed, and covered with wet clothes. All specimens were demolded 1day after casting. Thereafter, they were cured in water at a constant room temperature (30 °C). All specimens were dried in an oven at 50 ± 2 °C for 24 h before testing. Three specimens of each mix were prepared for flexural strength and compressive strength test.

2.4. Testing procedures

At the age of 28 days, the flexural strength and compressive strength tests were conducted. After the testing, thirty gram-shaped samples of each mix were collected randomly for measuring the porosity and pore size distribution (five gram-shaped samples were used for each test). An AUTOSCA-10 Mercury Intrusion Porosimetry (MIP), able to determine the distribution of pores from 2 to 5000 nm, was used for the measurement. The maximum pressure provided by this machine was 600 MPa. The contact angle and mercury surface tension used were 140° and 480.0 erg/cm^2, respectively.

Morphology and microstructure study was performed with Scanning Electron Microscopy (SEM). Three samples with a size about 1 cm × 1 cm × 1 cm were prepared for each mix at the age of 28 days after the compressive strength test. These samples were kept in alcohol till testing, and gold-coated before examination.

Analysis of infrared absorption spectra was conducted for cement paste by Nicolet 5DXC Fourier-transform infrared (FT-IR) spectrometer at the age of 90 days. Samples (100 g) of each mix were collected randomly, and then these samples were dried in an oven (50 ± 2 °C) for 24 h prior to chipping and grinding to pass a 45 μm sieve. Three samples of each mix for the testing were coated with a KBr pellet. IR spectra were obtained in the absorption mode with a resolution of 4 and 4000 cm^{-1} scans, respectively.

3. Results and discussion

3.1. Mechanical properties

Table 4 shows the results of both the compressive and flexural strengths of PCC, PCCF and PCNT after 28 days curing. It is observed that the use of carbon nanotubes can enhance both the compressive and flexural strengths of cement. The compressive strength increases up to 19%, while the flexural strength increases up to 25%. Although the addition of carbon fibers can

Table 4
Strengths of different mixes after 28 days curing

Mix	Compressive strength (MPa)	Flexural strength (MPa)
PCC	52.27 ± 1.4%	6.69 ± 1.5%
PCCF	47.51 ± 3.1%	8.14 ± 2.6%
PCNT	62.13 ± 2.3%	8.37 ± 2.1%

improve flexural strength, leading to an increase of 22%, the compressive strength is weakened.

Fig. 1 presents the typical compressive stress-strain curves of those cement composites. It is shown that the use of carbon nanotubes can also improve the deformation ability of cement, and lead to a significant increase of the failure strain. Though the untreated carbon fibers cause a decrease of the compressive strength, the failure strain is significantly increased.

3.2. Micrograph

Fig. 2 shows a micrograph of carbon nanotubes. It is observed that the diameter of nanotubes is between 20 and 40 nm and the length is about tens of microns (mm), resulting in an aspect ratio up to 100–1000. SEM analysis (Fig. 3) demonstrates that carbon nanotubes act as bridges across pores and cracks. Moreover, the treated carbon nanotubes are tightly wrapped by C–S–H. This indicates that high bonding strength between the carbon nanotubes and cement matrix is achieved. The untreated carbon fibers also act as a bridge across pores. However, the bonding force between the fibers and cement matrix may be low, since the fibers were not surface-untreated.

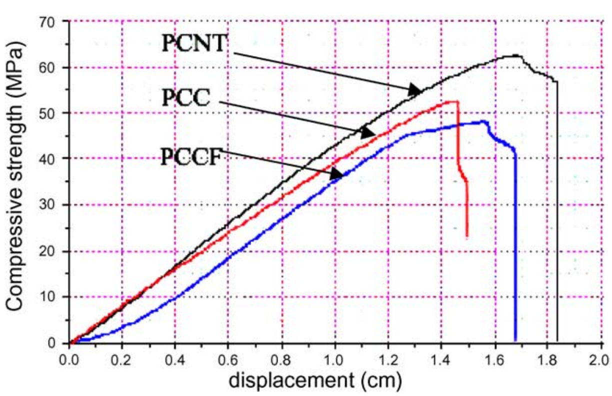

Fig. 1. Typical load–displacement curves of cement-matrix composites.

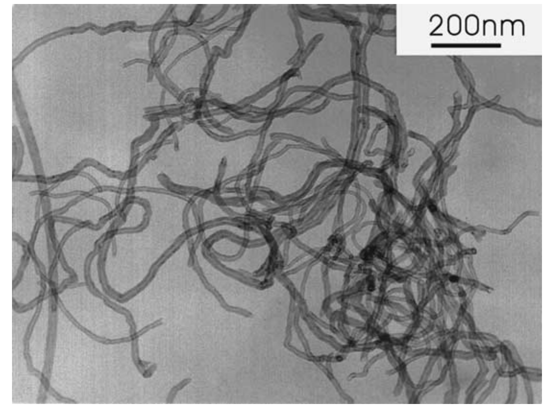

Fig. 2. TEM image of multi-wall carbon nano-tubes.

3.3. FT-IR spectra

The FT-IR spectrum of the treated carbon nanotubes is presented as curve "a" of Fig. 4. Since the surface of CNTs was oxidized with nitric acid, the peak at 1733 cm^{-1} is associated with C=O stretching of carboxylic acids, which occurs only for these CNTs with a stronger absorbance [20,21]. The peak at 1118 cm^{-1}, corresponding to the C-OH stretch mode, is observed much more intensely [20–23]. This indicates that the treatment by strong oxidizing acid causes the attachment of oxygen-containing groups (such as carboxylic and hydroxyl) to the surfaces of CNTs.

The spectrum of cement paste (PCC, curve d of Fig. 4) shows the appearance of bands at 3643, 3436, 1632, 1428, 980, 779, 678 and 457 cm^{-1}. Several workers studied the spectra of hydrated products of cement phase, and revealed that the bands observed in the range of 700 to 1200 cm^{-1} were lost when C_3S was hydrated [24]. The band at 980 cm^{-1} is due to the shifting of anhydrous phase and/or (Si–O) absorption bands of C_2SH_2, as well as C–S–H phases in the cement [24]. The absorption band due to (OH) of $Ca(OH)_2$ appears at 3643 cm^{-1}. The bands at 1632 and 3436 cm^{-1} are due to the presence of calcium sulphate in the form of ettringite (mono and trisulphate) in the cement. The bands at 779 cm^{-1} and 678 cm^{-1} are consistent with β-C_2S and α-C_2S

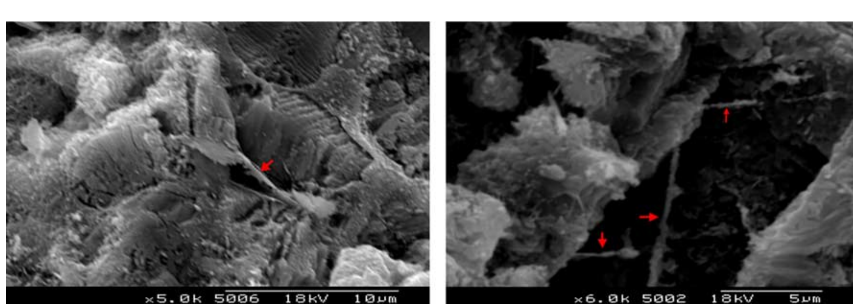

Fig. 3. Typical SEM image of carbon nanotubes cement matrix composites (Arrows: carbon nanotubes).

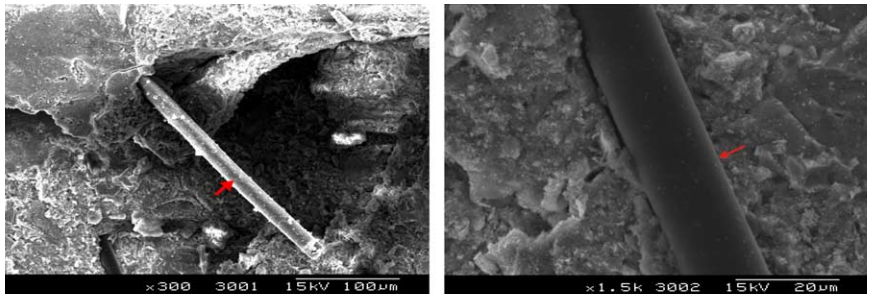

Fig. 4. Typical SEM image of carbon fibers cement matrix composites (Arrows: carbon fibers).

phases, respectively. The band at 1428 cm^{-1} shows the presence of carbon dioxide.

The spectrum of carbon nanotube-reinforced cement (PCNT, curve b of Fig. 4) shows the appearance of bands at 3434, 1756, 1630, 1426, 976, 875, 713 and 454 cm^{-1}. The peak at 1756 cm^{-1} (a positive shift of 22 cm^{-1} than the bands of CNTs) shows the possible existence of carboxylate, while the disappearance of band at 3643 cm^{-1} probably indicate that the interaction between the COOH or C-OH group attached to CNTs and $Ca(OH)_2$ occurred. The shape of curve b is obviously different from that of curve d from 1200 cm^{-1} to 400 cm^{-1} scans, indicating that the C–S–H phase in PCNT is different from one in PCC due to the interaction between the COOH or C-OH group attached to CNTs and the C–S–H phase of cement matrix.

The spectrum of carbon fiber-reinforced cement (PCCF, curve c of Fig. 4) shows the appearance of bands at 3643, 3430, 1633, 1424, 977, 773, 678 and 457 cm^{-1}, which are similar to these of PCC. This indicates that no new phase forms in PCCF.

3.4. Porosity and pore size distribution

The results of porosity test for PCC, PCCF and PCNT after 28 days' curing are shown in Table 5. It is observed that the use of carbon nanotubes decreases the porosity of cement and reduces the total pore volume. When containing 0.5% carbon nanotubes, PCNT has a total porosity of 10.8%, about 64% lower than that of PCC. Moreover, the pores with a size $d \geqslant 50$ nm in PCNT are 1.47%, about 82% lower than that of PCC. Those results suggest that carbon nanotubes act as the filler of voids. However, PCCF containing 0.5% carbon fibers has a total porosity of 23.4%, about 31% higher than that of PCC. The pores with a size $d \geqslant 50$ nm in PCCF are 9.89%, about 2.7 times higher than that of PCC.Fig. 5.

Mean volume pore radius (MVPR) refers to the minimum radius of pores when half of the total intrusion has taken place. At the age of 28 days, the MVPR value of PCC is about 13.77 nm, but the MVPR value of PCCF is much higher, about 22.33 nm. The MVPR value of PCNT is the lowest, about 7.7 nm.

Fig. 6 shows the variation of $\mathrm{d}V/\mathrm{d}\log R$ with pore radius. For PCC, $\mathrm{d}V/\mathrm{d}\log R$ increases with the decrease of pore radius, and reaches the highest value of 0.973 at the radius of 20 nm. Thereafter, $\mathrm{d}V/\mathrm{d}\log R$ decreases with the decrease of pore radius. For PCNT, a similar trend can be observed. However, the highest value of 0.973 appears in the much finer region of pores, at the radius of 10 nm. For PCCF, the highest value of $\mathrm{d}V/\mathrm{d}\log R$ is 1.39, which appears in the region of bigger pore size, at the radius of about 30 nm. These phenomena indicate that the presence of carbon nanotubes reduces the pore size, while the use of carbon fibers enlarges the pore size.

3.5. Discussion

Previous studies concluded that the addition of carbon filaments in cement matrix increase the tensile strength while significantly decrease the compressive strength of cement matrix composites [16]. Even if carbon filaments treated by using ozone gas and well dispersed by adding silica fume and methylcellulose, the compressive strength for carbon filaments reinforced-cement composites (48.6 MPa) is lower than that for plain cement (57.9 MPa). However, present results show that

Table 5
Porosity and mean radius of different mixes

Sample	Total intruded vol (ml/g)	Mean radius (nm)	Apparent density[a] (g/cm^3)	Porosity (2 nm < d < 5 µm) (%)	Porosity (d < 50 nm) (%)	Porosity ($d \geqslant 50$ nm) (%)
PCC	0.0737 ± 8.4%	15.23 ± 8.6%	2.41 ± 0.83%	17.76 ± 9.2%	15.09 ± 9.1%	2.67 ± 8.7%
PCCF	0.1097 ± 6.5%	20.72 ± 8.2%	2.13 ± 0.5%	23.37 ± 6.8%	13. 48 ± 5.6%	9.89 ± 6.6%
PCNT	0.0445 ± 7.6%	8.65 ± 5.7%	2.43 ± 2.0%	10.8 ± 4.7%	10. 13 ± 9.3%	1.47 ± 9.1%

[a] Measured by using Archimedes' method.

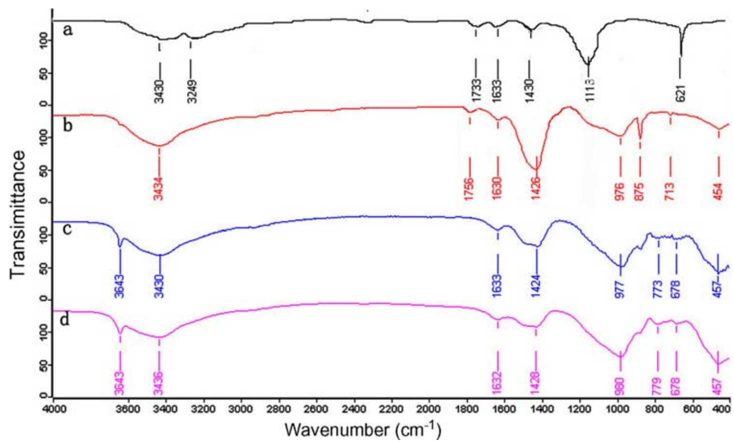

Fig. 5. Typical FT-IR spectra of 90-day cement matrix composites: (a) carboxylic carbon nanotubes, (b) carbon nanotubes-cement paste, (c) carbon fibers-cement paste, (d) plain cement paste.

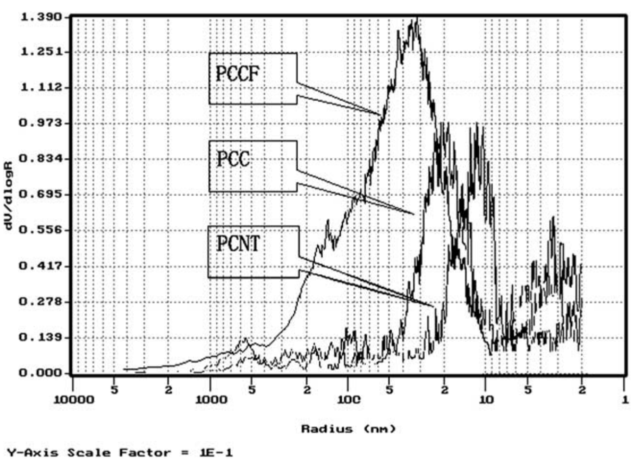

Fig. 6. Typical dV/d$\log R$ versus pore radius at 28 days.

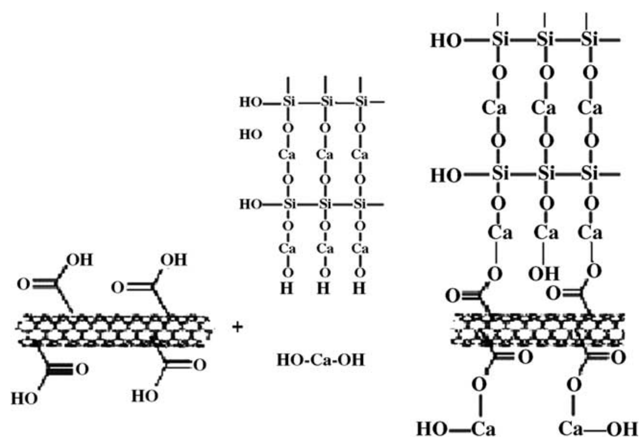

Fig. 7. Reaction scheme between carboxylated nanotube and hydrated production [$Ca(OH)_2$ and C–S–H] of cement.

the addition of the treated carbon nanotubes can improve the mechanical properties, and both the compressive and flexural strengths of cement composites are greatly increased. This increase is due to the improvement of the composite microstructure: (i) the occurrence of interfacial interaction between the treated nanotubes and cement. Due to the presence of carboxylic acid groups on the surfaces of carbon nanotubes when modified by using the mixed solution of sulfuric acid and nitric acid, chemical reactions take place between the carboxylic acid and the calcium silicate hydrate (C–S–H) or $Ca(OH)_2$, as demonstrated by SEM and FT-IR analysis. The general scheme of this reaction is illustrated in Fig. 7. The interaction leads to a strong covalent force on the interface between the reinforcement and matrix in the composites, and therefore increases the load-transfer efficiency from cement matrix to nanotubes. As a result, the mechanical properties of the composites are improved. (ii) The reduction of porosity in the cement composites. It is known that cement composites are porous materials, in which there exist large numbers of capillary pores and micro voids. The more the pores and voids, the lower the strengths. Due to the addition of carbon nanotubes, the porosity of the composites is reduced and the pore sizes are fined. Therefore, the composites become compacted, which will lead to the increase of strengths. (iii) The bridge-connecting effect of carbon nanotubes. As demonstrated by SEM analysis, carbon nanotubes act as the role of bridges across cracks and voids, and therefore guarantee the load-transfer in the case of tension. Certainly, the flexural strengths are enhanced.

4. Conclusion

The mechanical properties and microstructure of the cement composites containing carbon nanotubes have been investigated through experimental research. The carbon nanotubes used were surface-treated by using the mixed solution of sulfuric acid and nitric acid. It is found that the addition of carbon nanotubes to cement can greatly enhance its flexural and compressive strengths, as well as failure strain. This enhancement is due to the improvement of material microstructure. Interfacial interactions between surface-modified nanotubes and hydrations (such as C–S–H and calcium hydroxide) of cement will produce a high bonding strength, and increase the load-transfer efficiency from cement matrix to the reinforcement. Moreover, the addition of carbon nanotubes fines pore size distribution and decreases the porosity of cement composites. Therefore, the newly formulated composites become much more compacted. Furthermore, the bridge coupling effect of carbon nanotubes guarantees the load-transfer across voids and cracks in the case of tension.

References

[1] Philip B. The perfect nano-tubes. Nature 1996;382(18):207–8.

[2] Wong EW, Sheehan PE, Liebert CM. Nanobeam mechanics: elasticity, strength, and toughness, of nanorods and nanotubes. Science 1997;277:1971–3.

[3] Salvetat JP, Kuik AJ. Electronic and mechanical properties of carbon nanotubes. Adv Mater 1997;22:161–7.

[4] Tombler TW. Coupled electro-mechanical behavior of carbon nanotubes. Nature 2000;405:769–72.

[5] Iijima S. Helical microtubes of graphitic carbon. Nature 1991; 354(7):56–8.

[6] Wagner HD, Lourie O, Feldman Y, et al. Stress-induced fragmentation of multi-walled carbon nanotubes in a polymer matrix. Appl Phys Lett 1998;72(2):188–90.

[7] Qian D, Dickey EC, Andrews R, Rantell T. Load transfer and deformation mechanisms in carbon nanotube-polystyrene composites. Appl Phys Lett 2000;76(20):2868–70.

[8] Dong SR, Tu JP, Zhang XB. An investigation of the sliding wear behaviour of Cu-matrix composite reinforced by carbon nanotubes. Mater Sci Eng A 2001;313:83–7.

[9] Flahaut E, Peigney A, Laurent CH, Marlière Ch, Chastel F, Rousset A. Carbon nanotube-metal-oxide nanocomposites: microstructure electrical conductivity and mechanical properties. Acta Mater 2000;48:3803–12.

[10] Kuzumaki T, Miyazawa K, Ichinose H, Ito K. Processing of carbon nanotube reinforced aluminium composite. J Mater Res 1998;13:2445–9.

[11] Xu CL, Wei BQ, Ma RZ, Liang J, Ma XK, Wu DH. Fabrication of aluminum-carbon nanotube composites and their electrical properties. Carbon 1999;37:855–8.

[12] Peigney A, Laurent Ch, Flahaut E, Rousset A. Carbon nanotubes in novel ceramic matrix nanocomposites. Ceram Int 2000;26: 677–83.

[13] Ma RZ, Wu J, Wei BQ, Liang J, Wu DH. Processing and properties of carbon nanotubes–nano-SiC ceramic. J Mater Sci 1998;33:5243–6.

[14] Peigney A, Flahaut E, Laurent Ch, Chastel F, Rousset A. Aligned carbon nanotubes in ceramic-matrix nanocomposites prepared by high-temperature extrusion. Chem Phys Lett 2002;352:20–5.

[15] Balázsi Cs, Knóya Z, Wéber F, Biró LP, Arató P. Preparation and characterization of carbon nanotube reinforced silicon nitride composites. Mater Sci Eng C 2003;23:1133–7.

[16] Fu X, Chung DDL. Submicron-diameter-carbon-filament cement–matrix composites. Carbon 1998;36(4):459–62.

[17] Chung DDL. Comparison of submicron-diameter carbon filaments and conventional carbon fibers as fillers in composite materials. Carbon 2001;39:1119–25.

[18] Fu X, Chung DDL. Radio wave reflecting concrete for lateral guidance in automatic highways. Cement Concrete Res 1998; 28(6):795–801.

[19] Fu X, Chung DDL. Submicron carbon filament cement–matrix composites for electromagnetic interference shielding. Cement Concrete Res 1996;26(10):1467–72.

[20] Bae J, Jang J, Yoon S-H. Cure behavior of the liquid-crystalline epoxy/carbon nanotube system and the effect of surface treatment of carbon fillers on cure reaction. Macromol Chem Phys 2002; 203:2196–204.

[21] Ami E, Kuiyang J, Doug D, Rodney A, Schadler LS. Surface modification of multiwalled carbon nanotubes: toward the tailoring of the interface in polymer composites. Chem Mater 2003; 15:3198–201.

[22] Wang Z, Lou G, Chen JC, Xiao S, Wang Y. Carbon nanotubes as separation carrier in capillary electrophoresis. Electrophoresis 2003;24:4181–8.

[23] Jung D-H, Ko YK, Jung H-T. Aggregation behavior of chemically attached poly(ethylene glycol) to single-walled carbon nanotubes (SWNTs) ropes. Mater Sci Eng C 2004;24:117–21.

[24] Ghosh SN. Infrared spectroscopic study of cement and raw material. In: Cement and Concrete Science and Technology, vol. I, Part II, ABI Books, 404, New Delhi, 1992. p. 222–52.